higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

CARPENTRY AND ROOF WORK
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 The trusses carrying the support deck can be strengthened by nailing an extra member to the bottom chord. (1)

1.2 1.2.1 Will not
- The geyser is placed too close to the side of the roof structure
- The geyser is too far from the cold water supply and bathroom that will increase installation cost and electricity usage.

(Any applicable 2 × 1) (2)

1.2.2 Will not
- The geyser is too far from the Bathroom and kitchen that will increase the cost of electrical and plumbing installations
- There is no internal walls that can support the support deck

(Any applicable 2 × 1) (2)

1.2.3 Will
- The geyser is close to the cold water supply, and bathrooms positioned almost in the middle between the bathroom and kitchen points that will reduce installation cost
- Internal walls that can support the support deck structure
- Enough space for the installation of the geyser and fitments

(Any applicable 2 × 1) (2)

1.2.4 Will not
- The geyser is too far from the cold water supply and bathroom that will increase installation cost and electricity usage.
- The geyser is placed too close to the side of the roof structure

(Any applicable 2 × 1) (2)

1.3 Three (1)

1.4 For any geyser larger than 250 ℓ✓ and any truss with a span greater than 8 m, ✓you will need a design that has been signed off by an engineer.✓ (2)

1.5
- It is the wall that carries the load of the geyser – not the trusses.
- There is no added stress on the trusses.
- It is easy for workmen to work on the geyser should there be any repairs. (3)

[15]

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QUESTION 2

2.1
- Aesthetics: Improves the visual appearance of a room
- Thermal comfort: Helps limit temperature changes
- Strength: Adds strength when strong winds cause movement of the roof and walls
- Sound insulation: Prevents sound from getting in or out the room
- Dust: Prevents dust penetration into the building (Any 4 × 1) (4)

2.2
- Gypsum ceiling boards are wide and long – it covers wide spaces fast when installing the ceiling.
- Gypsum ceiling boards are cheap with less installation costs. (2)

2.3 A cornice is an ornamental moulding made of wood or gypsum✓ which runs along the top edge of a wall to decorate✓ or hide the edge of the ceiling where the ceiling and wall meet.✓ (3)

2.4
- Setting out: Assess the bottom chord and walls and check that the walls are parallel with the members to be installed. ✓If there is a difference of less than 15 mm between the walls, it can be adjusted by fanning the members slightly. ✓
- Start by loosely fixing a nail 10 mm away from the wall on the first bottom chord.✓ Do the same on the bottom chord at the other end of the room.✓ String a line between the two nails that will cross each of the remaining bottom chords in between.✓ Fix nails loosely into the rest of the bottom chords next to the line to form a straight row of nails parallel with the wall and all 10 mm away.✓ Choose now from a bundle the first member to be installed. Make sure that the first member is straight and remember to open only one bundle of members at a time to prevent warping.✓ Square off both ends of the first member with a mitre saw.✓ Place one end of the member in the corner with the grooved end against the nails.✓ By making use of the secret nailing method and 32 mm oval nails you can nail the first member to the bottom chord.✓ The first and last member each have two nails per bottom chord.✓ The one nail will be straight through the member that will be hidden by the cornice and the second nail on an angle that will be hidden by the next member (secret nailing method).✓ The next member can now be cut and squared off to fit.✓ The next member only needs the secret nailing method because the edge is held by the tongue of the previous member. If the room is larger than the length of the members, butt joints must be used.✓ When you fit the next row it is important to check the next member for defects and broken pieces stuck in the groove.✓ The same procedure must be followed when preparing the next member to be installed. To fit one member into the next, the member must be held on an angle and rotated into position while keeping the pressure on the member so that the groove slips over the tongue.✓ Do the same for all the other members. Place an offcut on the member to hammer the member in position. ✓
- Quality control by continuously checking if the tongue and groove boards are level. Stand back and look for straightness and also check for creep every ten lines. ✓

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- The last member should also rotate into position but leave 10 mm from the wall. It is most of the times required to rip the member to the required size with a rip saw or a table saw.✓ After all members are in place you can prepare and install the cornice.✓ (20)

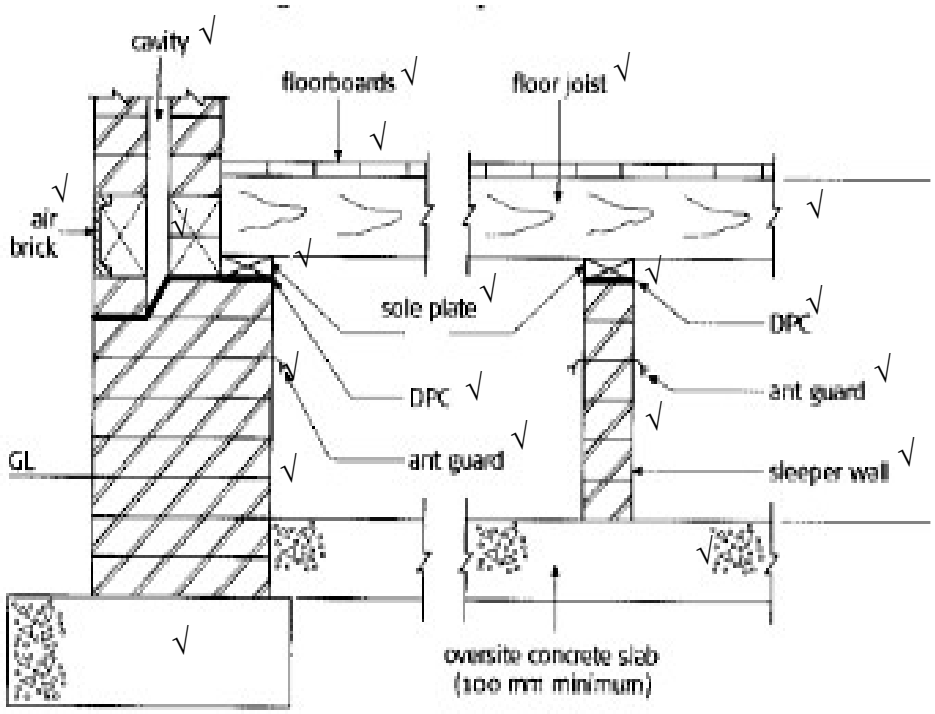
2.5
- It is not easily destroyed by moisture or humidity.√
- Fibreglass does not catch fire easily and irritates insects and rodents.√ (1)

[30]

QUESTION 3

3.1
- A suspended timber floor adds to the aesthetic value of the building.
- It regulates the temperature level inside the building.
- Suspended timber floors are easily repaired. (3)

3.2



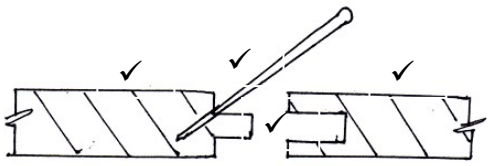
(20 × ½) (10)

3,3 3.3.1 Mezzanine floor is an intermediate (in-between) floor√ just above the ground level. √ (2 × ½) (1)

3.3.2 Sole plate is a horizontal member that is placed flat on the piers and outside walls.√ The floor joist rest on the sole plate.√ (2 × ½) (1)

3.3.3 Damp-proof membrane is a plastic membrane used beneath a casted concrete floor√ to prevent moisture rising from the ground into the floor structure.√ (2 × ½) (1)

3.4



(4)
[20]

QUESTION 4

4.1 4.1.1
- Measuring tape
- Pencil
- Try plane
- Circular saw
- Skill saw (Any applicable 2 × 1) (2)

4.1.2
- Mallet
- Mortice chisel (2)

4.1.3
- Marking knife
- Pencil
- Marking gauge (Any applicable 2 × 1) (2)

4.2 The weather strip is designed to prevent water from flowing under the door and into the house. (2)

4.3
- Locate the lock block in the door.
- The door is measured, marked.
- Cut the length and width, allowing for 2 mm clearance all around.
- Fit the door and trim /plane all the surfaces where the door is too big or too tight.
- The trimming and cutting must be on a slight angle (bevel cut) to prevent jamming.
- Fit the door and mark the positions for the butt hinges on the door and the doorframe where applicable.
- Accurately cut the marked recess/rebate for the butt hinges. Ensure that the butt hinges fit snugly in the recesses and are flushed with the surface of the edge of the door and the doorframe where applicable.
- Screw the hinges to the door in place and make sure that the screws and hinges are flushed with the door.
- Make a bevel cut to allow the diagonal of the thickness of the door to swing past the lock jamb.
- Check that the door swings well with no hinge-binding.
- Check that the door hangs on the correct hanging side.

(Any applicable 10 × 1) (10)

4.4 A latch bolt is a spring bolt, one edge which is bevelled√ so that when the door closes, the bolt is forced inwards against the bevelled side,√ pushing the spring inside.√ When in the fully closed position the bolt will spring back into a fixed notch or cavity.√ (4 × ½) (2)
[20]

QUESTION 5

5.1 A moulded trim is used to cover the gap✓ between the wall and cupboard.✓ (2)

5.2 A profile gauge is used to copy the profile of a surface to which the trim must be fitted✓ and then used to aid the marking of this profile onto the trim material to be cut.✓ (2)

5.3
- Meranti timber
- Pine timber
- Medium density fibre board
- Polywood (Any applicable 3 × 1) (3)

5.4 The shallow groove called a concave recess cut at the back of the wooden skirting is to prevent warping and to allow the top edge to seat tightly against the wall. (2)

5.5 Advantages:
- A large variety of colours and finishing are available.
- The boards are big and therefore minimises joints.
- It can be cut in accurate sizes that can provide a perfect fit.
- It is cheaper than solid wood.
- Chipboard surfaces that are smooth can be glued together.

(Any applicable 3 × 1)

Disadvantages:
- Chipboard is not water resistant and will crumble and disintegrate when getting wet.
- Melamine tends to chip or splinter when it is cut or damaged that will leave the chipboard exposed.
- Joints are not very strong and need additional support.
- Chipboard cannot be used for plinth constructions in kitchens and bath rooms. (Any applicable 3 × 1) (6)

[15]

TOTAL: 100

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